Drew University - CSCI 260 - Fall 2024
HW4 - LC-3 Assembly Language

HW Info.: Due on 10/25/2024 (F) by 11:59pm, 37 points
Submission: Submit as a single PDF file on Moodle.
Note: Please follow Drew University's Standard of Academic Integrity and refer to the course syllabus for late submission policy.

1. (15 points) Write an LC-3 program that utilizes R1 to count the number of 1s appeared in R0. For example, if we manually set R0 = 0001001101110000, then after the program executes, we should have R1 = #6.

   In order to get full credits, you need to give me (1) a flow chart on the program you design, and then (2) the assembly code of it, and lastly (3) the code converted to hexadecimal.

   Code:

```
; initialization
ADD     R2 R2 #15   ; add base-10 16 to loop counter R2
ADD     R2 R2 #1    ; for checking every bit
; loop body
BRz     #6          ; if loop counter is now 0, done and jump to HALT
AND     R0 R0 R0    ; get MSB in R0 via setting CC
BRzp    #1          ; if MSB not 1, skip next line
ADD     R1 R1 #1    ; otherwise, increment counter
ADD     R0 R0 R0    ; shift string to left by 1 spot
ADD     R2 R2 #-1   ; loop counter decrement
BRnzp   #-7         ; jump back to top of loop
; end of loop
HALT                ; program ends here
```

   Code in Hex:

```
x14AF
x14A1
x0406
x5000
x0601
x1261
x1000
x14BF
x0FF9
xF025
```

2. (12 points) Given that every register is initialized to zero, show the contents of the registers in hexadecimal after the following segment of code is fully executed.

```
0xA400   THIS1   LEA    R0, THIS1
0xA401   THIS2   LD     R1, THIS2
0xA402   THIS3   LDI    R2, THIS5
0xA403   THIS4   LDR    R3, R0, #2
0xA404   THIS5   .FILL  xA400
```

Ans:

- R0 loads the EA of the label THIS1, since the EA is xA400, R0=xA400.
- R1 loads the content within the label THIS2, and since the content there is "LD R1, #-1" (i.e., PC-1), R1 = 0010 001 111111111 = x23FF.
- R2 loads indirectly from THIS5, so the content of THIS5, xA400, is used as an address, and R2 is loaded with the content there, i.e., Mem[xA400] = "LEA R0, #-1"; hence, R2 = 1110 000 111111111 = xE1FF.

  However, note that at xA404, .FILL xA400 can also be interpreted "LDI R2, #0" which overwrites R2. Currently, PC is at xA405, so we are loading Mem[Mem[xA405]] = Mem[x0000] = x0000 assume our system didn't execute anything before. Therefore, R2 is filled with x0000; or if you verify in LC-3, you will end up with xFD00, as it is set by the system.
- R3 uses the content in R0, xA400, as an address and load from that address added by 2, i.e., Mem[xA400+#2] = "LDI R2, #1"; hence, R3 = 1010 010 000000001 = xA401.
- R4 – R7 are all x0000.

R0	xA400	R1	x23FF	R2	x0000	R3	xA401
R4	x0000	R5	x0000	R6	x0000	R7	x0000

3. (10 points) For the program below, describe what does it do using no more than 25 words, and find the value stored at the memory location labelled with RESULT after the program terminates.

```
          .ORIG   x3000
          LD      R2, ZERO
          LD      R0, M0
          LD      R1, M1
LOOP      BRz     DONE
          ADD     R2, R2, R0
          ADD     R1, R1, #-1
          BRnzp   LOOP
DONE      ST      R2, RESULT
          HALT
RESULT    .FILL   x0000
ZERO      .FILL   x0000
M0        .FILL   x0004
M1        .FILL   x0803
          .END
```

Ans: Multiply value stored at M0 location with value stored at M1 location, and store the product at RESULT location. In other words, multiply decimal numbers 4 with 2051 and store the product at x3009 location.

RESULT location contains the decimal value 8204 or hexadecimal value x200C.